

The Cyclic Spacetime Motion Hypothesis

A Four-Quadrant Geometric Extension of the Relativistic Proper-Time Relation

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September 2026

Status: Speculative theoretical hypothesis. The negative temporal-orientation branch introduced here is not part of standard special relativity.

Abstract

This paper develops a speculative geometric extension of the special-relativistic proper-time relation. Starting from $d\tau/dt = \sqrt{1 - v^2/c^2}$, a temporal-motion component $T \equiv c(d\tau/dt)$ is defined, yielding $T^2 + v^2 = c^2$ for one-dimensional signed spatial motion. The standard positive-root solution occupies an upper semicircle in the normalized $(v/c, T/c)$ state plane. The proposed Cyclic Spacetime Motion Hypothesis (CSMH) extends this representation through the negative root, producing a four-quadrant closed cycle. In the hypothesis, $T > 0$ denotes forward temporal orientation, $T = 0$ a null boundary, and $T < 0$ a proposed reversed temporal orientation. The construction never requires $|v| > c$: instead, the state vector rotates between temporal and spatial components while maintaining a fixed Euclidean state-space norm. A three-spatial-dimensional generalization produces a 3-sphere constraint $T^2 + v_x^2 + v_y^2 + v_z^2 = c^2$. The paper distinguishes carefully between identities inherited from special relativity, mathematical reparameterization, and the new physical postulate. Its principal unresolved problem is whether the negative temporal branch can be assigned a Lorentz-covariant physical meaning without contradicting causality, the timelike nature of massive worldlines, or the experimentally supported structure of special relativity.

1. Introduction

Special relativity relates coordinate time t , proper time τ , and ordinary speed v through the familiar time-dilation relation. For a massive particle following a timelike worldline, increasing spatial speed reduces the amount of proper time accumulated per unit coordinate time. In the limit $v \rightarrow c$ from below, $d\tau/dt \rightarrow 0$. Standard special relativity does not interpret this limit as a massive object rotating through the light barrier into a backward-time state; a massive particle cannot be accelerated through c within the theory.

The present work asks a narrower mathematical question: if the time-dilation identity is rewritten as a constant-norm relation between a signed spatial component and a temporal component, can its two real square-root branches be organized as a continuous closed geometry? The answer is yes at the level of algebra. The additional claim that the negative branch represents physically reversed temporal orientation is then introduced explicitly as a hypothesis rather than as a consequence of established relativity.

2. Relativistic Starting Point

For inertial motion in one spatial dimension, the proper-time relation is

$$d\tau/dt = \sqrt{1 - v^2/c^2} \quad (1)$$

Squaring Eq. (1) and rearranging gives

$$(d\tau/dt)^2 + v^2/c^2 = 1. \quad (2)$$

Define a quantity with dimensions of speed:

$$T \equiv c(d\tau/dt). \quad (3)$$

Equation (2) then becomes

$$T^2 + v^2 = c^2. \quad (4)$$

Equation (4) is an algebraic rewriting of the proper-time relation for the ordinary timelike branch. It should not be confused with the Minkowski spacetime metric: T is a derived state-space coordinate, not an additional ordinary velocity measured directly by an observer.

3. Circular Parameterization

Normalize the two components by c and introduce an angular state parameter θ :

$$v/c = \sin \theta, \quad T/c = \cos \theta. \quad (5)$$

Then

$$\sin^2 \theta + \cos^2 \theta = 1 \quad (6)$$

and Eq. (4) is satisfied identically. The parameter θ is not assumed to be a physical spatial angle. It labels position in the proposed two-component state space.

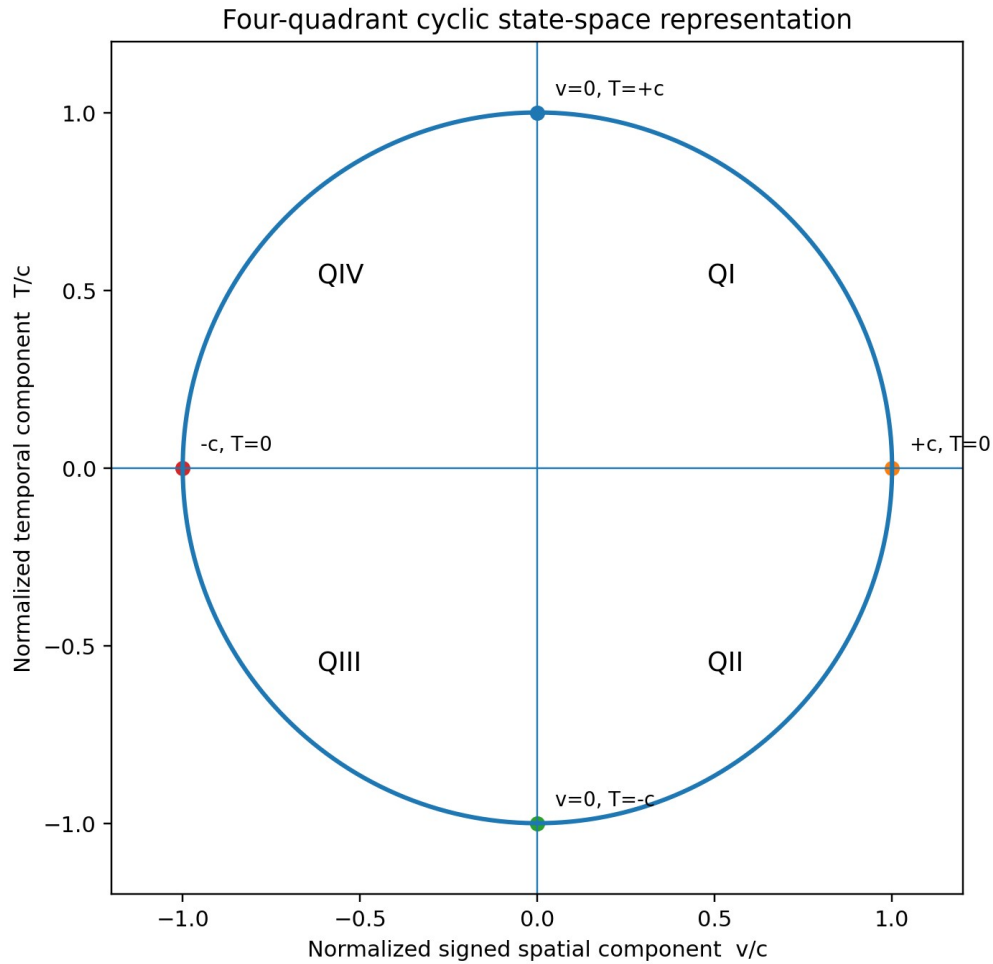


Figure 1. The proposed four-quadrant cycle in normalized state space. The horizontal axis is signed spatial motion; the vertical axis is the derived temporal component.

4. Four-Quadrant Hypothesis

Quadrant	θ range	v	T	Proposed interpretation
I	$0 \rightarrow \pi/2$	$0 \rightarrow +c$	$+c \rightarrow 0$	Forward temporal orientation; increasing forward spatial motion
II	$\pi/2 \rightarrow \pi$	$+c \rightarrow 0$	$0 \rightarrow -c$	Negative temporal orientation; forward spatial motion decreases to zero
III	$\pi \rightarrow 3\pi/2$	$0 \rightarrow -c$	$-c \rightarrow 0$	Negative temporal orientation; backward spatial motion increases in magnitude
IV	$3\pi/2 \rightarrow 2\pi$	$-c \rightarrow 0$	$0 \rightarrow +c$	Forward temporal orientation; backward spatial motion returns to zero

The full proposed cycle is therefore $(v,T) = (0,+c) \rightarrow (+c,0) \rightarrow (0,-c) \rightarrow (-c,0) \rightarrow (0,+c)$. The two states with $v = 0$ are distinguished by temporal orientation: $(0,+c)$ and $(0,-c)$. Thus no imaginary zero is required; $0i = 0$ carries no additional information.

5. Relation to the Positive and Negative Roots

From Eq. (4), the two real branches are

$$T = \pm c\sqrt{(1 - v^2/c^2)}, \quad |v| \leq c. \quad (7)$$

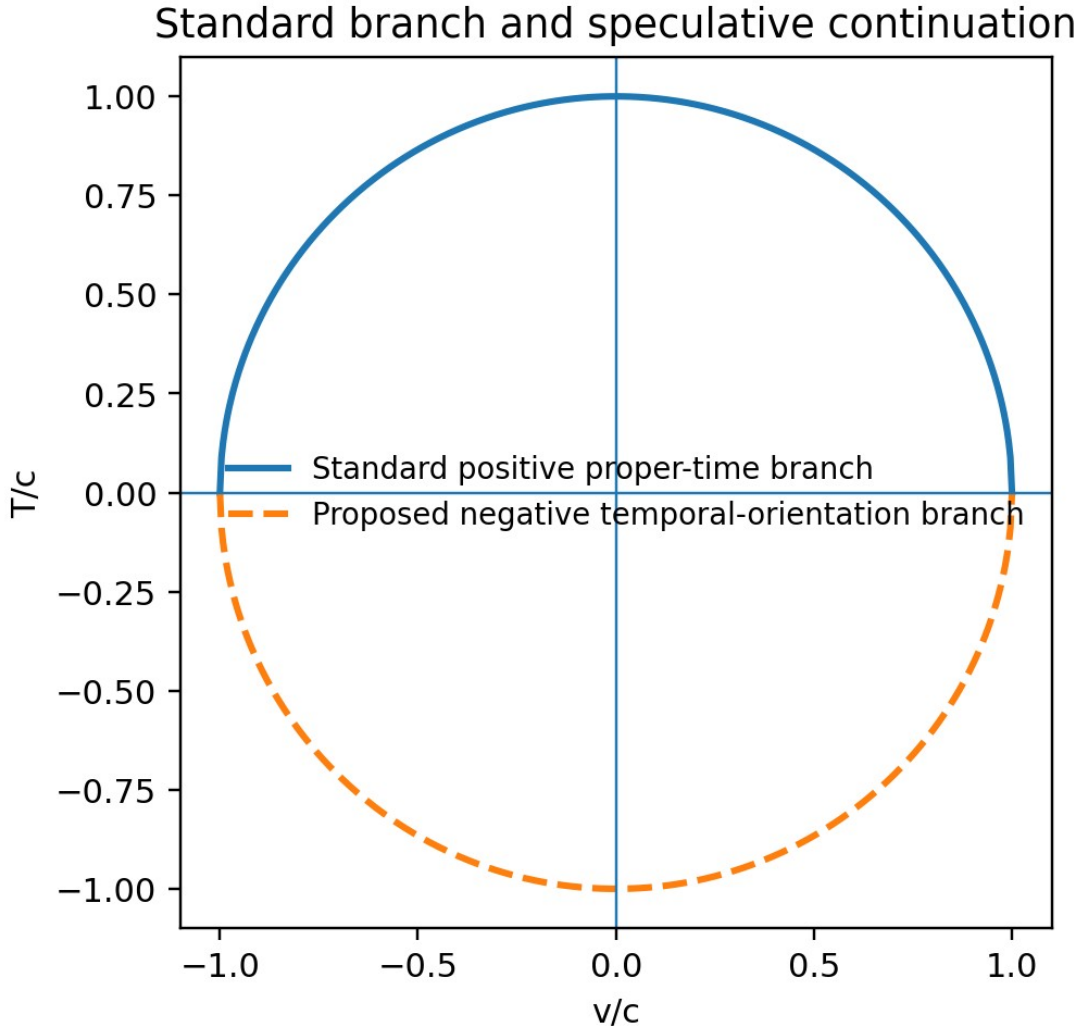


Figure 2. The positive proper-time branch used for ordinary future-directed timelike motion and the speculative negative branch.

In conventional relativistic particle kinematics, proper time along a future-directed timelike worldline increases, and the positive branch is used. CSMH makes its central new postulate here: the negative branch is treated as a distinct temporal orientation and is joined continuously to the positive branch at the null points $v = \pm c$, $T = 0$.

6. Why Superluminal Speed Is Not Required

An earlier way to motivate the construction is to substitute $v > c$ into Eq. (1), which makes dt/dt imaginary. That algebraic continuation is valid as complex algebra, but it describes a spacelike interval rather than a timelike massive worldline in standard special relativity. The four-quadrant construction presented here takes a different route: it retains $|v| \leq c$ everywhere and changes the sign of T after a null boundary.

$$|v| \leq c \quad \text{for all } \theta. \quad (8)$$

This makes the model mathematically cleaner: the proposed reversal is encoded in the temporal component rather than by assigning a spatial speed greater than c . However, the transition through $v = \pm c$ remains physically problematic for a massive particle, because standard relativistic dynamics requires unbounded energy as v approaches c .

7. Three Spatial Dimensions

Let the ordinary spatial velocity be the vector $\vec{v} = (v_x, v_y, v_z)$, with $v^2 = v_x^2 + v_y^2 + v_z^2$. Equation (4) generalizes to

$$T^2 + v_x^2 + v_y^2 + v_z^2 = c^2. \quad (9)$$

After normalization by c , this is the equation of a unit 3-sphere S^3 embedded in a four-dimensional Euclidean state space. The circle of Figure 1 is a great-circle cross-section obtained when the direction of the spatial velocity is held fixed. This observation matches the intended picture of three mutually orthogonal spatial components plus one temporal component, while avoiding the incorrect identification of a single imaginary axis with all three spatial dimensions.

8. Comparison with Minkowski Spacetime

The geometric circle in CSMH must not be identified with the invariant geometry of special relativity. Minkowski spacetime has a Lorentzian, not Euclidean, metric. With the $(+, -, -, -)$ convention, the infinitesimal interval is

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2. \quad (10)$$

For timelike motion $ds^2 > 0$; for lightlike motion $ds^2 = 0$; and for spacelike separation $ds^2 < 0$. Lorentz transformations preserve this interval and generate hyperbolic rather than ordinary circular rotations between space and time coordinates. CSMH instead constructs a Euclidean circle in a derived velocity/proper-time state space. Any physical theory based on CSMH would therefore need to demonstrate how this state-space circle coexists with Lorentz invariance.

9. Causality and Temporal Orientation

The interpretation $T < 0$ as backward temporal evolution is the strongest and least established part of the hypothesis. Mathematically, changing the sign of $d\tau/dt$ reverses the orientation of the worldline parameter relative to coordinate time. That fact alone does not demonstrate that an observer can physically enter such a branch, send information to their own past, or complete the proposed cycle.

A successful extension would have to specify a dynamical law for θ , explain what happens to energy and momentum at $T = 0$, preserve or deliberately modify Lorentz covariance, and resolve causal-loop problems. Without such a mechanism, the negative branch should be understood as a geometric postulate rather than an experimentally supported prediction.

10. Testable Structure and Falsifiability

To progress from a kinematic picture to a physical theory, CSMH would need at least one observable consequence not already contained in special relativity. Candidate questions include whether a finite-energy transition across $T = 0$ is predicted, whether the sign of T couples to measurable energy, momentum, decay, or clock rates, and whether the theory predicts departures from Lorentz symmetry. If no operational distinction can be defined between the proposed negative branch and a mere reversal of parameterization, then the construction remains a mathematical representation rather than a new physical model.

11. Limitations

- The identity $T^2 + v^2 = c^2$ follows from the standard proper-time equation only after defining $T = c(d\tau/dt)$; T is not an independently established physical velocity.

- The closed lower semicircle is not predicted by standard special relativity. Its interpretation as reversed time is an added postulate.
- A massive object cannot reach $v = c$ using finite energy in standard relativistic dynamics, so the proposed quadrant transitions lack a known physical mechanism.
- The Euclidean state-space circle is not the Minkowski spacetime metric and must not be used as a replacement for Lorentzian geometry without further derivation.
- The model currently supplies kinematics but no field equations, interaction law, quantum description, or experimentally discriminating prediction.

12. Conclusion

The Cyclic Spacetime Motion Hypothesis reorganizes the relativistic proper-time identity into a constant-norm state-space relation. With $T = c(dt/d\tau)$, the ordinary one-dimensional relation becomes $T^2 + v^2 = c^2$ and admits the angular parameterization $(v, T) = (c \sin\theta, c \cos\theta)$. Standard future-directed motion occupies the positive- T branch. The proposed extension joins this branch to a negative- T branch to form a complete four-quadrant cycle while retaining $|v| \leq c$. In three spatial dimensions the corresponding constraint is an S^3 state space. The construction is mathematically consistent as a parameterization, but its interpretation as a physical cycle through forward and backward temporal orientations is not established by special relativity. The next theoretical task is therefore not further geometric visualization but the construction of a Lorentz-consistent dynamical law capable of deciding whether the negative branch represents new physics or only a reparameterization.

References

- [1] Einstein, A. (1905). On the Electrodynamics of Moving Bodies. *Annalen der Physik*, 17, 891–921.
- [2] Minkowski, H. (1909). Space and Time. *Physikalische Zeitschrift*, 10, 104–111. (Lecture delivered 1908.)
- [3] Rindler, W. (2006). *Relativity: Special, General, and Cosmological* (2nd ed.). Oxford University Press.
- [4] Taylor, E. F., & Wheeler, J. A. (1992). *Spacetime Physics: Introduction to Special Relativity* (2nd ed.). W. H. Freeman.
- [5] Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman.